

Project Goal

The goal of this project is to explore the relationships between inflation, unemployment, interest rates, and 10-year Treasury yields in the U.S. from 2000–2026.

Using Python and Pearson correlation analysis, the project examines how these economic indicators are related and whether their relationships change across different economic periods.

Data Source & Attribution

This dataset was compiled from economic data available through the Federal Reserve Bank of St. Louis (FRED), covering 2000–2026.

Series used:

- CPIAUCSL: Consumer Price Index
- UNRATE: Unemployment Rate
- DFF: Federal Funds Effective Rate
- DGS10: 10-Year Treasury Yield

The original data sources include the U.S. Bureau of Labor Statistics and the Board of Governors of the Federal Reserve System. The data were accessed through FRED, then cleaned, transformed, and combined for analysis and educational purposes.

Please refer to the original FRED series for source information and applicable usage terms.

Why I Downloaded the CPI Data from 1999

Although the analysis covers 2000–2026, I downloaded the CPI data beginning in 1999.

The reason is that the CPI was originally provided as an index, and I needed to calculate year-over-year inflation. Since the data is monthly, calculating inflation for a given month requires the CPI value from the same month 12 months earlier.

For example, to calculate inflation for January 2000, I need the CPI value from January 1999.

Downloading the additional year of data therefore allowed me to calculate the 2000 inflation values instead of losing the entire first year to missing values.

Understanding the CPI Index

The Consumer Price Index is reported as an index, not directly as an inflation percentage.

The CPI uses 1982–1984 = 100 as its reference period. This means that 100 represents the average price level during that reference period. It does not mean that inflation was 100%.

For example, if the CPI is 170 today, that means the overall price level is substantially higher than it was during the 1982–1984 reference period.

Transforming CPI into Inflation

For this project, I was interested in the inflation rate, rather than the raw CPI index.

I therefore used the monthly CPI data to calculate the year-over-year percentage change. This compares each month's CPI with the CPI from the same month one year earlier.

This transformation gives me an Inflation variable expressed as a percentage, which is more meaningful for the economic analysis in this project.

The original CPI index was retained as part of the data preparation process, while the newly calculated Inflation variable was used for the main analysis.